7 TROUBLESHOOTING

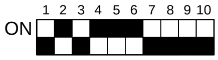
[1] Principal Parts

(1) Pressure sensor

1) Check for failure by comparing the sensing pressure according to the high pressure/low pressure pressure sensor and the pressure gauge pressure.
Turn on switches 1, 3, 5, 6 (High) and 2, 4, 5, 6 (Low) of the digital display select switch (SW1) as shown below, and the sensor pressure of the high pressure/low pressure sensors is displayed digitally by the light emitting diode LD1.

High pressure

Low pressure



1 In the stopped condition, compare the pressure readings from the gauge and from the LD1 display.
(a) If the gauge pressure is 0~0.098MPa the internal pressure is dropping due to gas leakage.
(b) If the pressure according to the LD1 display is 0~0.098MPa there is faulty contact at the connector, or it is disconnected. Proceed to 4.
(c) If the pressure according to the LD1 display is 3.14MPa or higher, proceed to 3.
(d) If other than (a), (b) or (c), compare the pressure readings during operation. Proceed to 2.

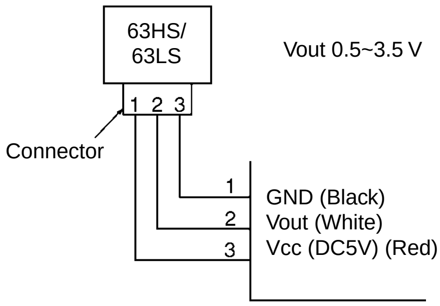
2 Compare the pressure readings from the gauge and from the LD1 display while in the running condition.
(a) If the difference between the two pressures is within 0.098MPa, both the affected pressure sensor and the main MAIN board are normal.
(b) If the difference between the two pressures exceeds 0.098MPa, the affected pressure sensor is faulty (deteriorating performance).
(c) If the pressure reading in the LD1 display does not change, the affected pressure sensor is faulty.

3 Disconnect the pressure sensor from the MAIN board and check the pressure according to the LD1 display.
(a) If the pressure is 0~0.098MPa on the LD1 display, the affected pressure sensor is faulty.
(b) If the pressure is 3.14MPa (in the case of the low pressure sensor, 0.98MPa) or higher, the MAIN board is faulty.

4 Disconnect the pressure sensor from the MAIN board and short out the No. 2 and No. 3 pins of the connector (63HS, 63LS), then check the pressure by the LD1 display.
(a) If the pressure according to the LD1 display is 3.14MPa (in the case of the low pressure sensor, 0.98MPa) or higher, the affected pressure sensor is faulty.
(b) If other than (a), the MAIN board is faulty.

2) Pressure sensor configuration.
The pressure sensors are configured in the circuit shown in the figure at right. If DC 5 V is applied between the red and black wires, a voltage corresponding to the voltage between the white and black wires is output and this voltage is picked up by the microcomputer. Output voltages are as shown below.

High pressure	0.1 V per 0.098MPa
Low pressure	0.3 V per 0.098MPa



※Connector connection specifications on the pressure sensor body side.
The connector's pin numbers on the pressure sensor body side differ from the pin numbers on the main circuit board side.

	Sensor body side	MAIN board side
Vcc	Pin 1	Pin 3
Vout	Pin 2	Pin 2
GND	Pin 3	Pin 1

(2) Solenoid valve (SV1~6, SV71~73)

Check if the control board's output signals and the operation of the solenoid valves match.
Setting the self-diagnosis switch (SW1) as shown in the figure below causes the ON signal of each relay to be output to the LED's.
Each LED shows whether the relays for the following parts are ON or OFF. When a LED lights up, it indicates that the relay is ON.

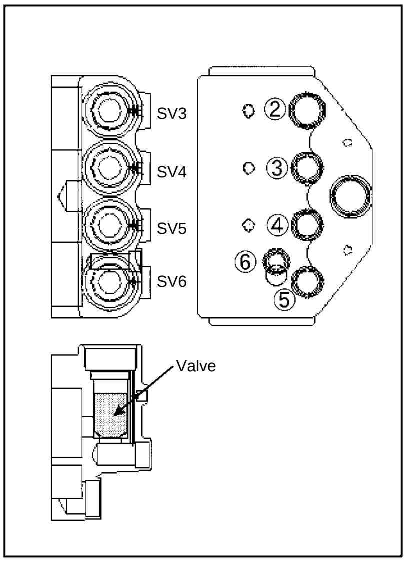
SW1	LED							
	1	2	3	4	5	6	7	8
				SV1	SV2	SV3	SV4	
	SV5	SV6	SV71	SV72	SV73			

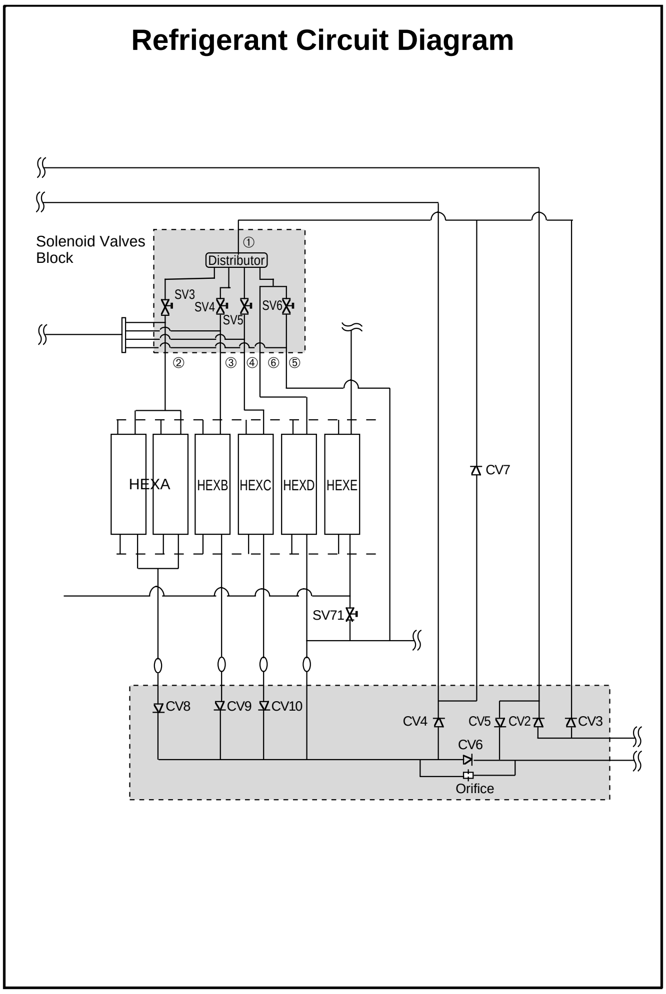
1) In the case of SV1 (Bypass Valve)
   (a) When the compressor starts, SV1 is ON for 4 minutes, so check operation by whether the solenoid valve is emitting an operating noise.
   (b) Changes in the operating condition by solenoid valve operation can be confirmed by the temperature of the bypass circuit and the sound of the refrigerant.

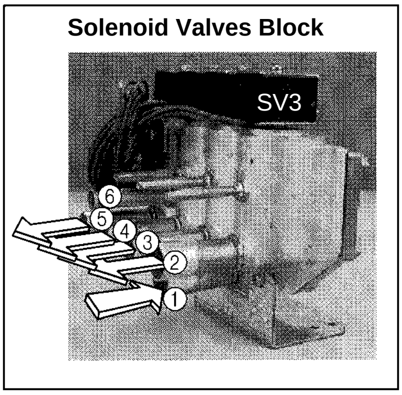
2) In the case of SV2 (Bypass)
   (a) SV2 goes ON in accordance with the rise in the high pressure in the cooling mode and heating mode, so check its operation by the LED display and the operating noise emitted by the solenoid valve.
   (Conditions during operation: See **Control of Heat Source Unit.**)
   (b) Changes in the operating condition by solenoid valve operation can be confirmed by the temperature of the bypass circuit and the sound of the refrigerant.

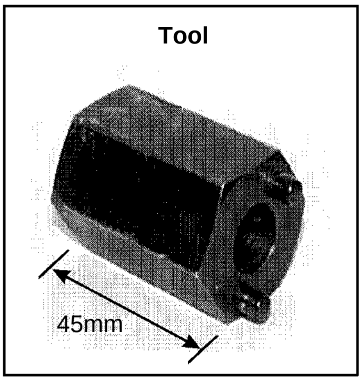
3) SV3~6, SV71~73 (Control of heat exchanger capacity)
   (a) Operations can be confirmed by LED display and operating sound of solenoid valve, because one or more of SV3~5, SV71 are turned on depending on conditions during cooling-only operations.
   (b) Operation can be confirmed by LED display and operating sound of solenoid valve, because all of SV3~5, SV73 are turned on during heating-only operations.
   (c) Operations can be confirmed by LED display and operating sound of solenoid valve, because one or more of SV3~6, SV71~73 are turned on depending on conditions during cooling-principal and heating-principal operations.

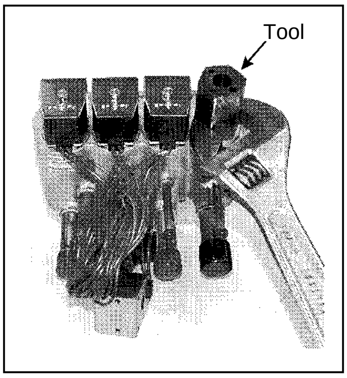
(d) The refrigerant flow is as following figure. Hot gas (high pressured) flows in cooling mode and cool gas/liquid (low pressured) flows in heating mode. Please refer to the Refrigerant Circuit Diagram.
And, ON/OFF of Solenoid valve is depends on the amount of running indoor units, ambient temperature and so on. So please check by LED Monitor Display.
The SV coil is taken off, then it is possible to open caps and check plungers. But the special tool which is on the Service Parts List is needed.

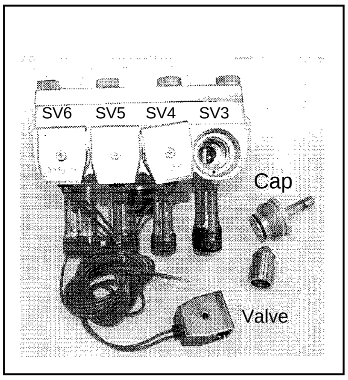










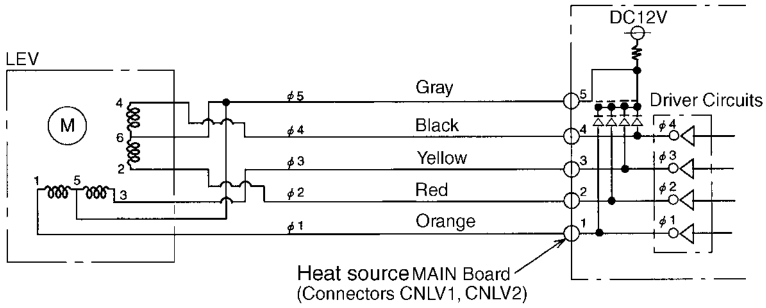


❋Closed torque : 1.3N·m

(3) LEV

• Heat source unit

The valve opening angle changes in proportion to the number of pulses.
(Connections between the heat source unit's MAIN board and SLEV, LEV2)



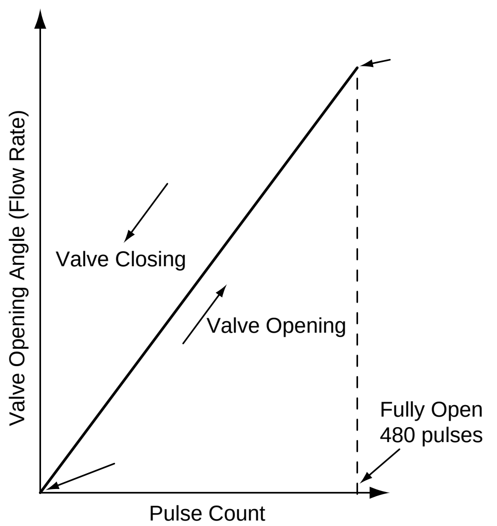
Pulse signal output and valve operation

Output (phase)	Output states							
	1	2	3	4	5	6	7	8
ø1	ON	OFF	OFF	OFF	OFF	OFF	ON	ON
ø2	ON	ON	ON	OFF	OFF	OFF	OFF	OFF
ø3	OFF	OFF	ON	ON	ON	OFF	OFF	OFF
ø4	OFF	OFF	OFF	OFF	ON	ON	ON	OFF

Output pulses change in the following orders when the
Valve is Closed 1→2→3→4→5→6→7→8→1
Valve is Open 8→7→6→5→4→3→2→1→8

※1. When the LEV opening angle does not change, all the output phases are off.
2. When the output is out of phase or remains ON continuously, the motor cannot run smoothly, but move jerkily and vibrates.

LEV valve closing and valve opening operations



※ When the power is switched ON, a 520 pulse valve opening signal is output to make sure the valve's position, so that it is definitely at point A. (The pulse signal is output for approximately 17 seconds.)

※ When the valve operates smoothly, there is no sound from the LEV and no vibration occurs, but when the valve is locked, it emits a noise.

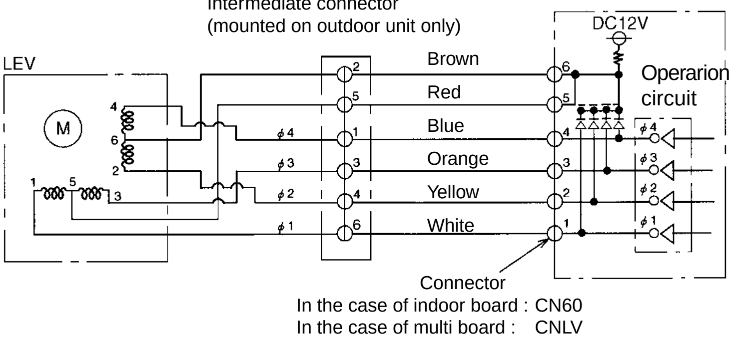
※ Whether a sound is being emitted or not can be determined by holding a screwdriver, etc. against it, then placing your ear against the handle.

※ If there is liquid refrigerant inside the LEV, the sound may become lower.

• **BC controller and indoor unit**

① LEV receives pulse signal from microcomputer, and operates valve with stepping motor.
② Valve opening changes in proportion to the number of pulses.

Connection of microcomputer circuit board and LEV



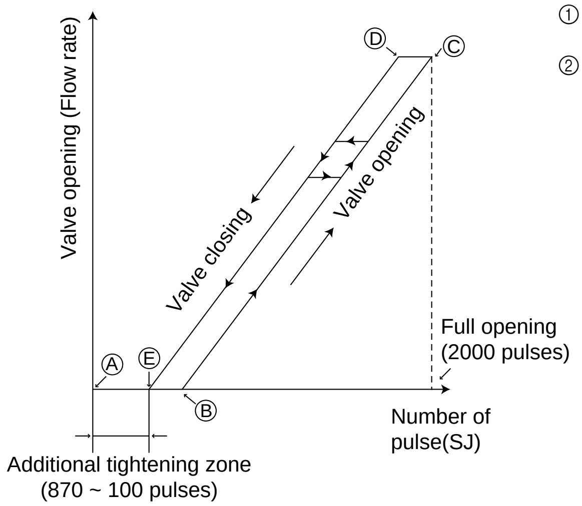
Note:
Pay attention to colors of lead wires because numbers of intermediate connectors are different from those of circuit board side connectors.

Pulse signal output and valve operations

Output (phase) No.	Output states			
	1	2	3	4
ø1	ON	OFF	OFF	ON
ø2	ON	ON	OFF	OFF
ø3	OFF	ON	ON	OFF
ø4	OFF	OFF	ON	ON

① Valve open : Output pulse changes in order of 1→2→3→4→1.
Valve close : Output pulse changes in order of 4→3→2→1→4.
② All output phases are turned OFF when LEV opening does not change.
③ In case output phase is lacking or kept “ON,” motor can not rotate smoothly, generating ticking sound and vibration.

Closing and opening operations of valve

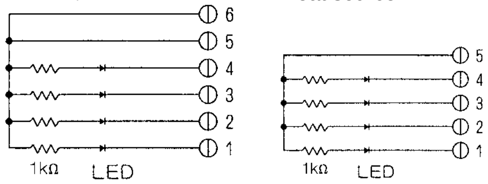
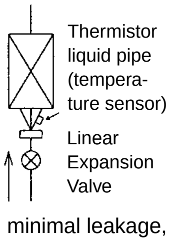


① When turning on power source, issue valve closing signal of 2,200 pulses, so that valve opening is located at point Ⓐ.
② When valve runs smoothly, no sound or vibration is generated from LEV. However, big sound is observed when valve opening changes from point Ⓔ to Ⓐ or valve is locked.
(Sound generation can be identified from the bundle of screwdriver attached to the valve.)

• **Judgment methods and likely failure mode**

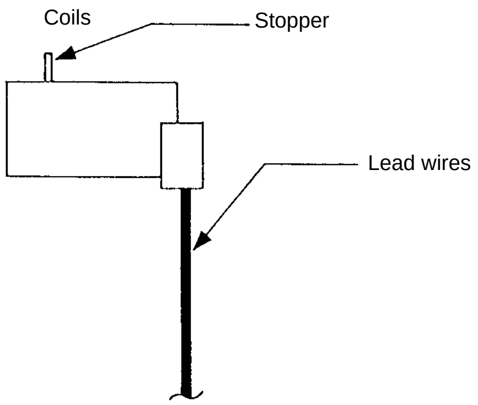
Caution:

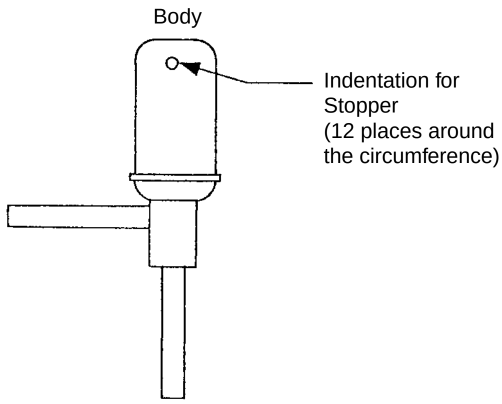
The specifications of the heat source unit (heat source LEV) and indoor unit (indoor LEV) differ. For this reason, there are cases where the treatment contents differ, so follow the treatment specified for the appropriate LEV as indicated in the right column.

Failure Mode	Judgment Method	Treatment	Affected LEV
Microcomputer driver circuit failure	① Disconnect the control board connector and connect the check LED as shown in the figure below. Indoor, BC controller Heat source  When the base power supply is turned on, the indoor LEV outputs pulse signals for 10 seconds, the heat source LEV outputs pulse signals for 17 seconds, and BC controller outputs pulse signals for 10-20 seconds. If the LED does not light up, or lights up and remains on, the driver circuit is abnormal.	In the case of driver circuit failure, replace the control board.	Indoor BC controller Heat source
LEV mechanism is locked.	① If the LEV is locked up, the drive motor turns with no load and a small clicking sound is generated. Generation of this sound when the LEV is fully closed or fully open is abnormal.	Replace the LEV.	Indoor BC controller Heat source
The LEV motor coils have a disconnected wire or is shorted.	Measure the resistance between the coils (red - white, red - orange, brown - yellow, brown - blue) using a tester. They are normal if the resistance is within 150Ω ± 10%.	Replace the LEV coils.	Indoor BC controller
	Measure the resistance between the coils (gray - orange, gray - red, gray - yellow, gray - black) using a tester. They are normal if the resistance is within 46Ω ± 3%.	Replace the LEV coils.	Heat source
Fully closed failure (valve leaks)	① If you are checking the indoor unit's LEV, operate the indoor unit's blower and the other indoor units in the cooling mode, then check the piping temperatures (liquid pipe temperatures) of the indoor units by the operation monitor through the heat source unit's control board. When the fan is running, the linear expansion valve is fully closed, so if there is leakage, the temperature sensed by the thermistor (liquid pipe temperature sensor) will become low. If the temperature is considerably low compared to the remote control's intake temperature display, it can be judged that there is a fully closed failure. In the case of minimal leakage, it is not necessary to replace the LEV if there are no other effects. 	If there is a large amount of leakage, replace the LEV.	Indoor BC controller
Faulty wire connections in the connector or faulty contact.	① Check for pins not fully inserted on the connector and check the colors of the lead wires visually. ② Disconnect the control board's connector and conduct a continuity check using a tester.	Check the continuity at the places where trouble is found.	Indoor BC controller Heat source

• **Heat source LEV (SLEV) coil removal procedure (configuration)**

As shown in the figure, the heat source LEV is made in such a way that the coils and the body can be separated.

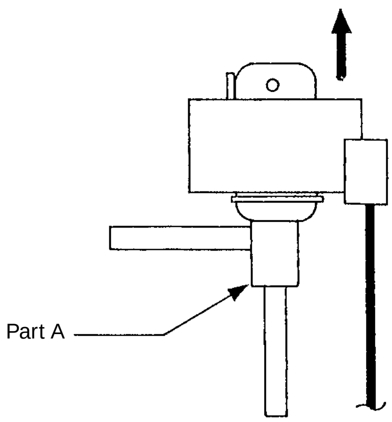




<Removing the coils>

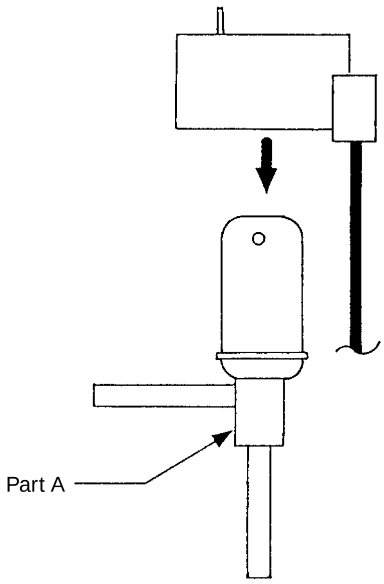
Fasten the body tightly at the bottom (Part A in the figure) so that the body will not move, then pull out the coils toward the top. If they catch on the stopper and are difficult to take out, turn the coils left and right until the stoppers are free from the stopper indentations, then pull the coils out.

If you take out the coils only without gripping the body, undue force will be applied to the piping and the pipe may be bent over, so be sure to fasten the body in such a way that it will not move.



<Installing the coils>

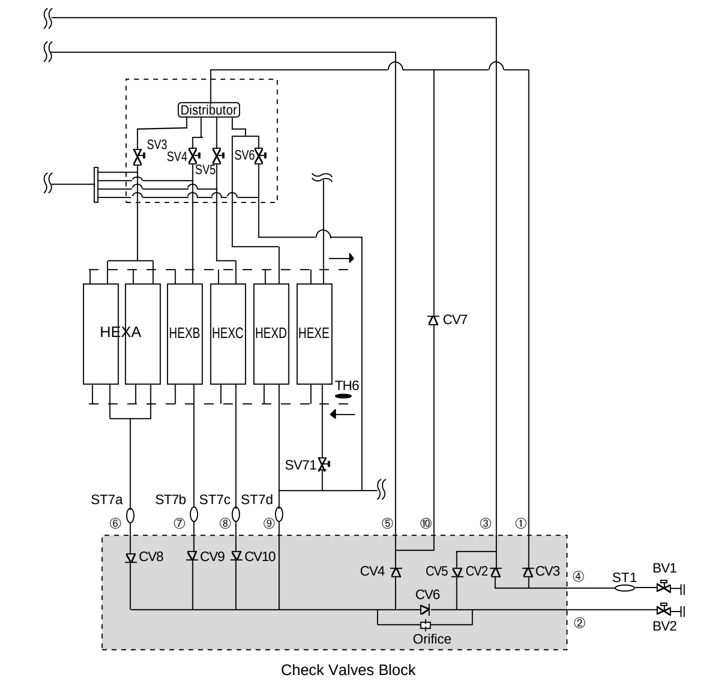
Fasten the body tightly at the bottom (Part A in the figure) so that the body will not move, then insert the coils from the top, inserting the coils' stopper securely in one of the indentations on the body. (There are four indentations for the stopper on the body around its circumference, and it doesn't matter which indentation is used. However, be careful not to apply undue force to the lead wires or twist them around inside the body.) If the coils are inserted without gripping the body, it may exert undue force on the piping, causing it to become bent, so be sure to hold the body firmly so that it won't move when installing the coils.



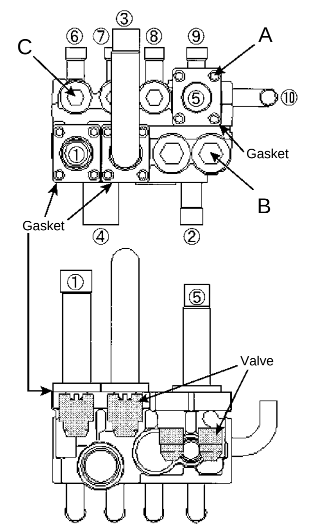
(4) Check valves block

The refrigerant flow in the pipe ⑥, ⑦, ⑧ and ⑨ are depend on ON/OFF of the SV3, 4, 5 and 6.
Please confirm by LED monitor display.
You can open the cap of valve A, B and C, but 3 types of hexagon socket screw keys. The size is as follows.

Refrigerant Circuit Diagram

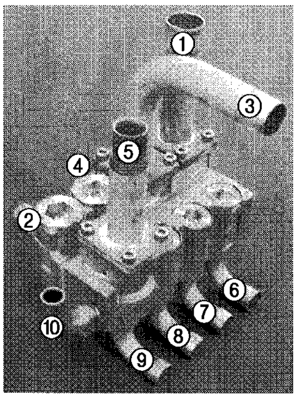


Check Valves Block

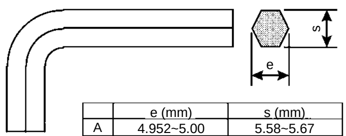


⁂ Closed torque : A : 1.7kg·m (0.17N·m)
B : 20kg·m (2.0N·m)
C : 13kg·m (1.3N·m)

Check valves Block

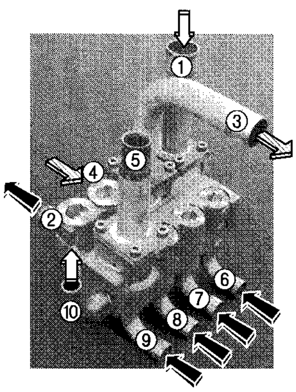


Hexagon socket screw key

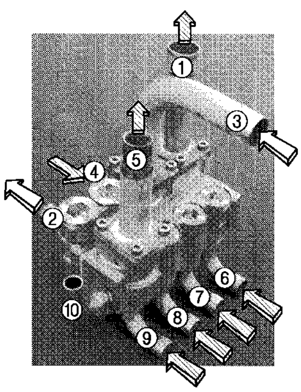


	e (mm)	s (mm)
A	4.952~5.00	5.58~5.67
B	18.87~19.00	21.32~21.63
C	13.89~14.00	15.70~15.93

Flow of ref. in cooling



Flow of ref. in heating



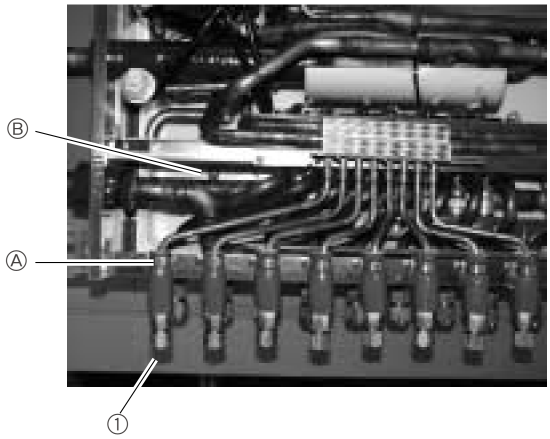
High pressure gas
High pressure liquid
Low pressure gas/liquid

Solenoid valves (SVA, SVB, SVC)
Coordination signals output from the board and solenoid valve operations.

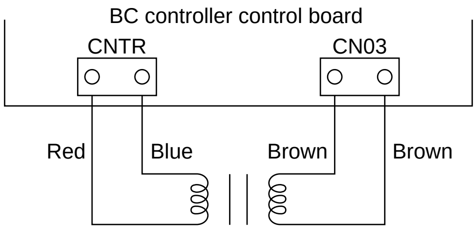
Note 1 : (SVA, SVB, SVC)
SVA, SVB and SVC are turned on and off in accordance with operation mode.

Mode / Branch port	Cooling	Heating	Stopped	Defrosting
SVA	ON	OFF	OFF	OFF
SVB	OFF	ON	OFF	OFF
SVC	ON	OFF	OFF	OFF

Note 2 : (SVA, SVB, SVC)
Measure temperature of piping on either side of SVA ①-Ⓐ
Measure temperature of piping on either side of SVB ①-Ⓑ



BC controller transformer



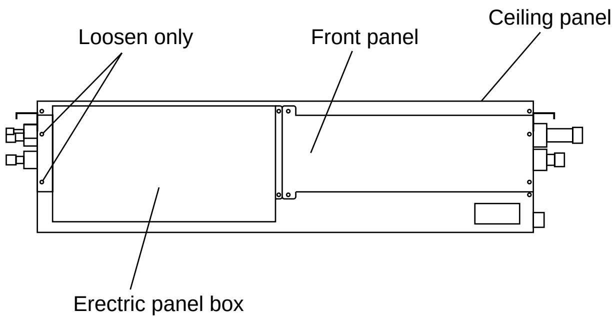
	Normal	Malfunction
CNTR(1)-(3)	Approximately 90Ω	Open or shorted
CN03(1)-(3)	Approximately 1.7Ω	

* Disconnect the connector before measurement.

[2] BC Controller Disassembly Procedure

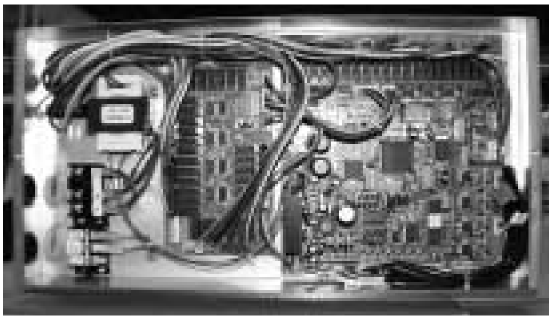
(1) Service panel

Be careful on removing heavy parts.

Procedure	Illustrations
1. Remove the two screws securing the electric panel box. Loosen the two screws securing the electric panel box, and then remove the box. 2. Remove the four screws securing the front panel and then remove the panel. 3. Remove the nine screws securing the ceiling panel and then remove the panal.	

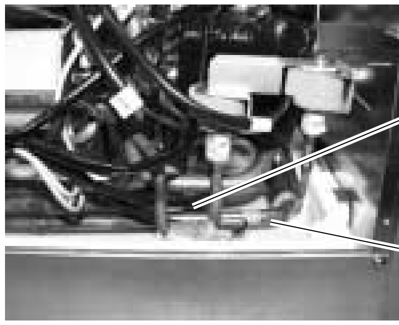
(2) Control Box

Be careful on removing heavy parts.

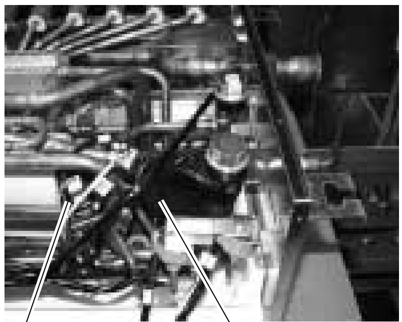
Procedure	Photos
Removing the two screws that secures the electric panel box cover provides access to the controller board and all of the relay board for checking. So it is not necessary to work according to above 2.	

(3) Thermistor (Liquid and gas piping temperature detection)

Be careful when removing heavy parts.

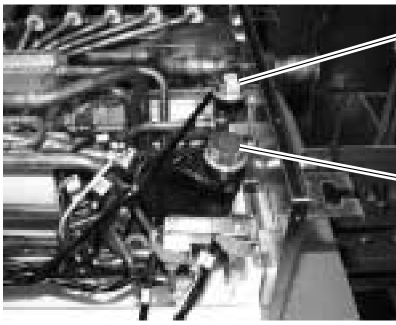
Procedure	Photos
1. Remove the front panel ① Use the procedure under (1)-1.2.3 to check TH11, TH12, TH15, and TH16. 2. Disconnect the piping sensor lead from the controller panel. ① TH11 - TH12 (CN10) ② TH15, TH16 (CN11) 3. Pull the temperature sensor from the temperature sensor housing and replace it with a new sensor. 4. Connect the temperature sensor lead securely to the controller board.	 

(4) Pressure Sensor

Procedure	Photos
1. Remove the front panel. ① Use the procedure under (1)-1.2 to check PS1 and PS3. 2. Disconnect the connector of the applicable pressure sensor from the controller board and insulate the connector. ① Liquid pressure sensor (CNP1) ② Intermediate pressure sensor (CNP3) 3. Install a new pressure sensor at the location shown in the photograph, and plug the connector into the controller board. Important ① In the case of gas leakage from the pressure sensor, take actions to fix the leak before performing the above procedure.	

(5) LEV

Be careful on removing heavy parts.

Procedure	Photos
1. Remove the service panel. See (1)-1.2.3 2. Replace the applicable LEV. Important! ① When performing the above procedure, be sure to allow for enough service space in the ceiling area for welding. ② When conditions require, the unit can be lowered from the ceiling before starting work.	

(6) Solenoid Valve Coil

Procedure	Photos
1. Remove the service panel. See (1)-1.2.3 2. Disconnect the connector of the applicable solenoid valve. 3. Remove the solenoid valve coil. ① SVA, SVB solenoid valve coils can be serviced from the maintenance port. SVC can serviced from the back if service space is available in the back. To remove the back panel, remove the four screws that secure it.	

(7) Transmission Power Circuit (30 V) Check Procedure

If "◎" is not displayed by the remote control, investigate the points of the trouble by the following procedure and correct it.

No.	Check Item	Judgment	Response
1	Disconnect the transmission line from TB3 and check the TB3 voltage.	DC24~30 V	Check the transmission line for the following, and correct any defects. Broken wire, short circuit, grounding, faulty contact.
		Except the above-mentioned	to No. 2
2	Check if the following connectors are disconnected in the outdoor unit's control box. MAIN Board: CNS1, CNVCC3, CNVCC4 INV Board: CNVCC2, CNVCC4, CNL2, CNR, CNAC2	Connector disconnected	Connect the connectors as shown on the electric wiring diagram plate.
		Except the above-mentioned	to No. 3
3	Disconnect the wires from CNVCC3 on the Main board and check the voltage between pins 1 and 3 on the wire side of the CNVCC3. Tester ⊕ ..... 1 pin Tester ⊖ ..... 3 pin	DC24~30 V	Check the wiring between CNS1 and TB3 for the following, and correct any defects. Broken wire, short circuit, grounding, faulty contact. If there is no trouble, replace the Main board.
		Except the above-mentioned	to No. 4
4	Disconnect the wiring from CNVCC2 on the INV board and check the voltage between pins 1 and 3 of CNVCC2. Tester ⊕ ..... 1 pin Tester ⊖ ..... 3 pin	DC24~30 V	Check the wiring between CNVCC2 and CNVCC3 for the following, and correct any defects. Broken wire, short circuit, grounding, faulty contact.
		Except the above-mentioned	to No. 5
5	Disconnect the wiring from CNL2 on the INV board, and check the resistance at both ends of choke coil L2.	0.5~2.5Ω	to No. 6
		Except the above-mentioned	Replace choke coil L2.
6	Disconnect the wiring from CNR on the INV board, and check the resistance at both ends of R7.	19~25Ω	to No. 7
		Except the above-mentioned	Replace R7.
7	Check the resistance at both ends of F01 on the INV board.	0Ω	to No. 8
		Except the above-mentioned	Replace F01
8	Check the voltage between pins 1 and 3 of CNAC2 on the INV board.	AC198~264 V	Replace the INV board.
		Except the above-mentioned	to No. 9
9	Check the voltage between L2 and N on power supply terminal block TB1.	AC198~264 V	Check the wiring to CNAC2 for the following and correct any defects. Broken wire, faulty contact.
		Except the above-mentioned	Check the power supply wiring and base power supply, and correct any defects.

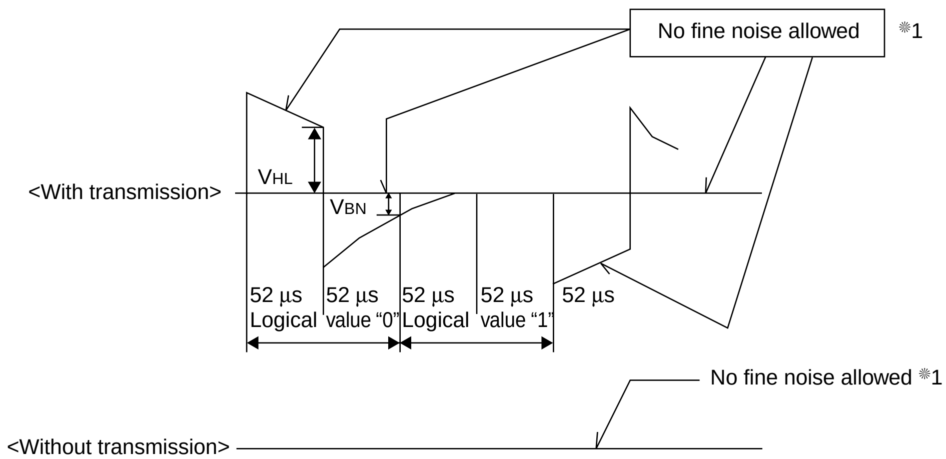
(8) Investigation of transmission wave shape/noise

Control is performed by exchanging signals between heat source unit, indoor unit and remote controller by M-NET transmission. If noise should enter into the transmission line, the normal transmission will be hindered causing erroneous operation.

1) Symptom caused by the noise entered into transmission line

Cause	Erroneous operation	Error code
Noise entered into transmission line	Signal changes and is misjudged as the signal of other address.	6600
	Transmission wave shape changes to other signal due to noise.	6602
	Transmission wave shape changes due to noise, and can not be received normally thus providing no reply (ACK).	6607
	Transmission can not be made continuously due to the entry of fine noise.	6603
	Transmission can be made normally, but reply (ACK) or answer can not be issued normally due to noise.	6607 6608

2) Method to confirm wave shape



Check the wave shape of transmission line with an oscilloscope to confirm that the following conditions are being satisfied.

① The figure should be 104µs/bit ± 1%.
② No finer wave shape (noise) than the transmission signal (52µs ± 1%) should be allowed. ❋1
③ The sectional voltage level of transmission signal should be as follows.

Logic value	Transmission line voltage level
0	V_{HL} = 2.0V or more
1	V_{BN} = 1.3V or less

❋1 However, minute noise from the DC-DC converter or inverter operation may be picked up.

3) Checking and measures to be taken

(a) Measures against noise

Check the items below when noise can be confirmed on wave shape or the error code in the item 1) is generated.

Items to be checked		Measures to be taken
Checking for wiring method	① Wiring of transmission and power lines in crossing.	Isolate transmission line from power line (5cm or more). Never put them in a same conduit.
	② Wiring of transmission line with that of other system in bundle.	Wire transmission line isolating from other transmission line. Wiring in bundle may cause erroneous operation like crosstalk.
	③ Use of shield wire for transmission line (for both indoor unit control and centralized control).	Use specified transmission wire. Type : Shield line CVVS/CPEVS Wire diameter : 1.25mm^2 or more
	④ Repeating of shield at the repeating of transmission line with indoor unit.	The transmission line is wired with 2-jumper system. Wire the shield with jumper system as same for transmission line. When the jumper wiring is not applied to the shield, the effect against noise will be reduced.
	⑤ Are the unit and transmission lines grounded as instructed in the INSTALLATION MANUAL?	Connect to ground as shown in the INSTALLATION MANUAL.
Check for earthing	⑥ Earthing of the shield of transmission line (for indoor unit control) to heat source unit.	One point earthing should be made at heat source unit. Without earthing, transmission signal may be changed as the noise on the transmission line has no way to escape.
	⑦ Arrangement for the shield of transmission line (for centralized control).	For the shield earth of the transmission line for centralized control, the effect of noise can be minimized if it is from one of the heat source units in case of the group operation with different refrigerant systems, and from the upper rank controller in case the upper rank controller is used. However, the environment against noise such as the distance of transmission line, the number of connecting sets, the type of connecting controller, and the place of installation, is different for the wiring for centralized control. Therefore, the state of the work should be checked as follows. a) No earthing • Group operation with different refrigerant systems One point earthing at heat source unit • Upper rank controller is used Earthing at the upper rank controller b) Error is generated even though one point earth is being connected. Earth shield at all heat source units. Connect to ground as shown in the user's manual.

(b) When the wave height value of transmission wave shape is low, 6607 error is generated, or remote controller is under the state of "HO."

Items to be checked		Measures to be taken
	⑧ The farthest distance of transmission line is exceeding 200m.	Confirm that the farthest distance from heat source unit to indoor unit/remote controller is less than 200m.
	⑨ The types of transmission lines are different.	Use the transmission wire specified. Type of transmission line : Shield wire CVVS/CPEVS Wire dia. of transmission line : 1.25mm^2 or more
	⑩ No transmission power (30V) is being supplied to the idoor unit or the remote control.	Refer to "Transmission Power Supply (30V) Circuit Check Procedure."
	⑪ Faulty indoor unit/remote controller.	Replace heat source unit circuit board or remote controller.

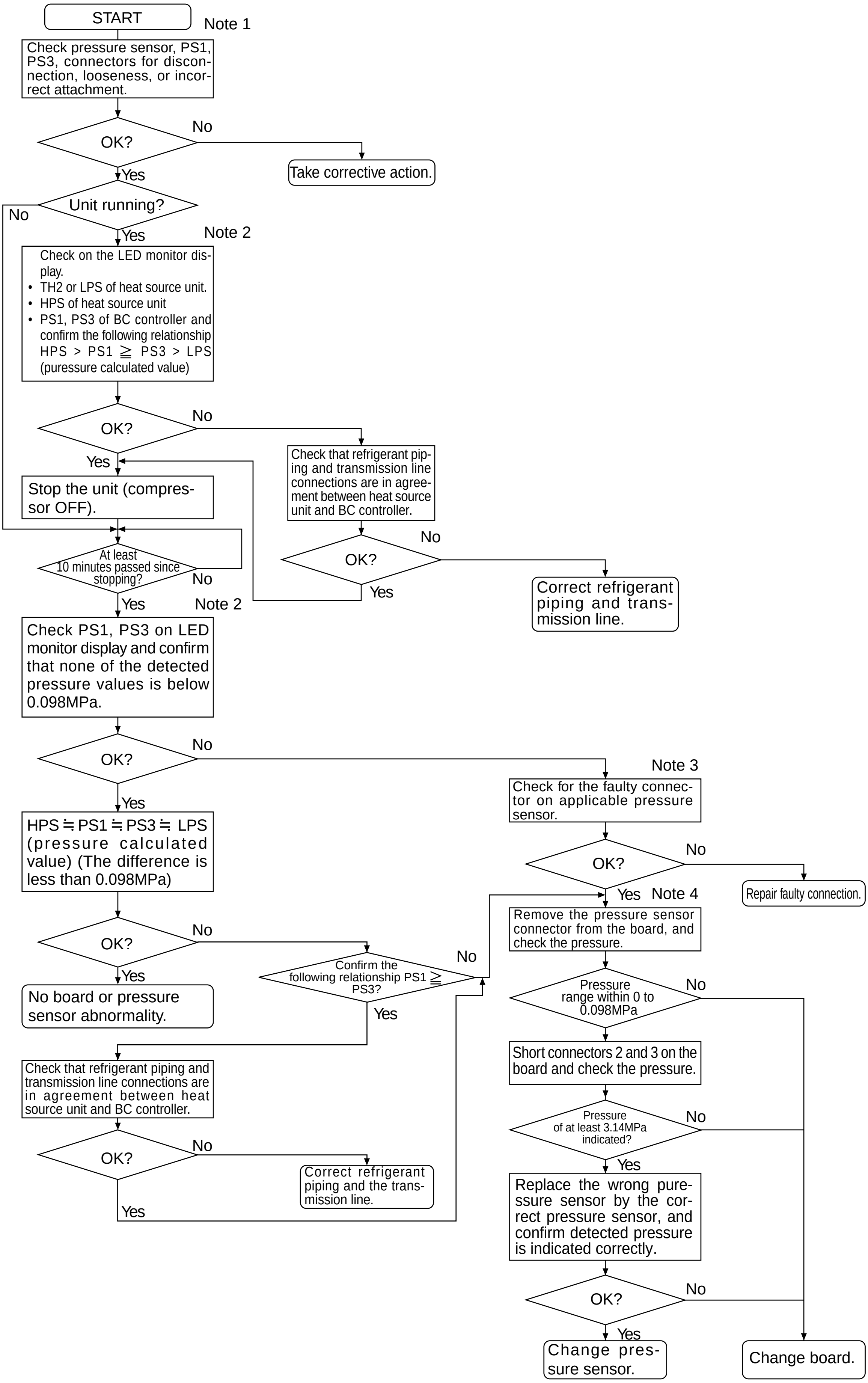
(9) Troubleshooting at breaker tripping

Check items	Measures to be taken
$\boxed{1}$ Check the breaker capacity.	The breaker's capacity should be proper.
$\boxed{2}$ Check the a short circuit or grounding in the electrical system other than the inverter.	Correct any defects.
$\boxed{3}$ Check the resistance between terminals on the terminal block TB1A for power source.	Check each part inside the inverter power circuit (resistance, megohm or the like). a) Diode stack Refer to "Troubleshooting of diode stack." b) IPM Refer to "Troubleshooting of IPM." c) Rush current protection resistor d) Electromagnetic contactor e) DC reactor ❋ For c) ~ e), refer to "Individual Parts Failure Judgement Methods."
① 0 ~ several ohms or improper megohm value	
$\boxed{4}$ Checking by powering again.	
① Main power source circuit breaker tripping	
② No display of remote controller	
$\boxed{5}$ Operational check by operating air conditioner	
① Normal operation without breaker tripping.	a) As there is a possibility of instantaneous short circuit generated, find the mark of the short circuit for repair. b) When a) is not applicable, the compressor may be faulty.
② Breaker tripping	The ground fault of inverter output/compressor can be supposed. Disconnect the wiring to the compressor and check the insulation resistance of the following parts with a megger. a) Compressor terminals. b) Inverter output.

(10) Troubleshooting the major components of the BC controller

1) Pressure sensor

Pressure sensor troubleshooting flow



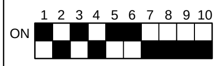
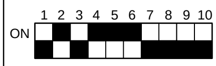
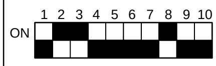
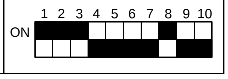
Note 1 :

- Symptoms of incorrect connection of BC controller pressure sensor to the board

Symptom						
Cooling-only	Cooling-principal		Heating-only		Heating-principal	
Normal	Insufficient cooling.	SC11 large SC16 small △ PHM < 0	Warm indoor SC small. Warm in-door thermo ON especially noise.	SC11 small SC16 small △PHM < 0	Insufficient heating Warm indoor SC small Warm indoor thermo ON especially noise	SC11 large SC16 small △ PHM < 0

Note 2 :

- Check using LED monitor display switch (heat source unit MAIN board SW1)

Measured Data	Signal	SW1 Setting
High pressure	HPS	
Low pressure	LPS	
BC controller pressure (liquid measurement)	PS1	
(intermediate)	PS3	

Note 3 :

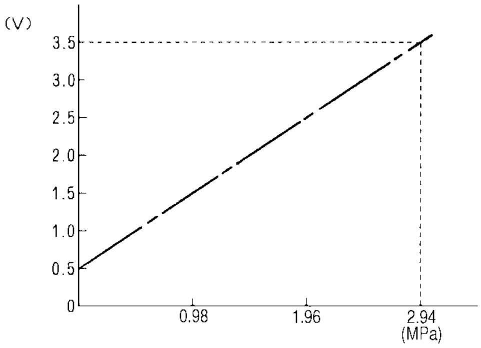
- Check CNP1 (liquid measurement) and CMP3 (intermediate) connectors on BC controller board for disconnection or looseness.

Note 4 :

- With the sensor of the applicable connector removed from the board, use the LED monitor display switch (Note 1) to check the pressure value.

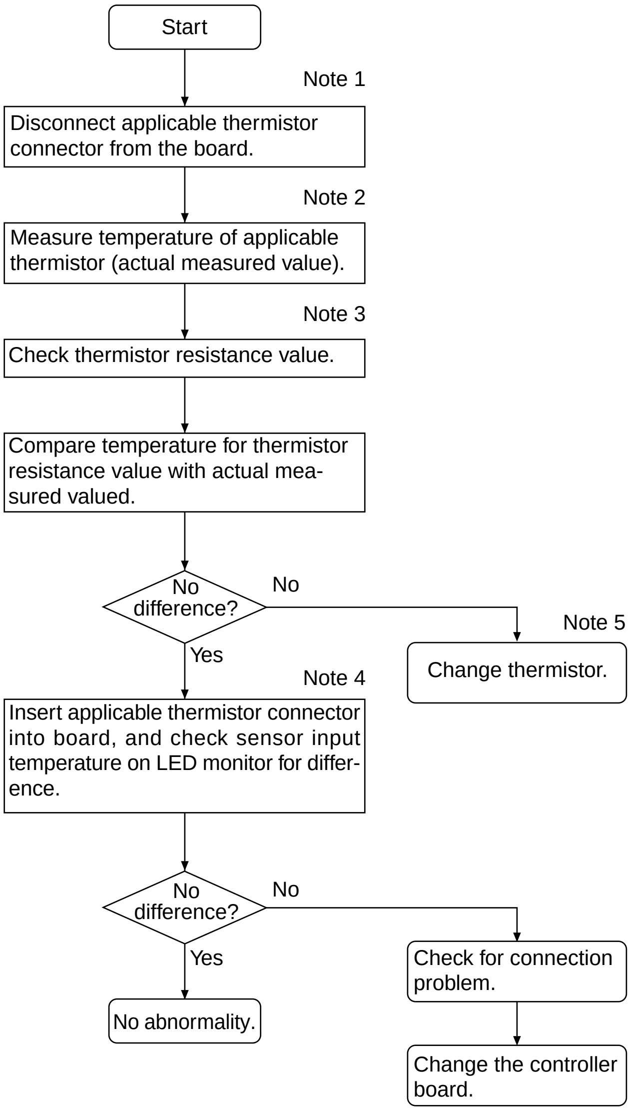
Pressure Sensor Replacement Precaution

(Pressure sensor output voltage)



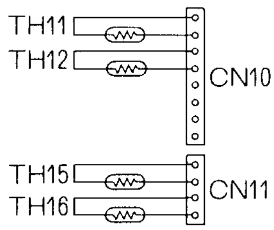
2) Temperature sensor

Thermistor troubleshooting flow



Note 1 :

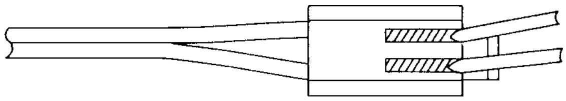
- Board connector CN10 corresponds to TH11 through TH12, while connector CN11 corresponds to TH15 through TH16. Remove the applicable connector and check the sensor for each number.



Note 2, 3 :

1. Pull the sensor connector from the I/O board. Do not pull on the lead wire.
2. Measure resistance using a tester or other instrument.
3. Compare measured values with values on the graph below. A value within a range of ±10% is normal.

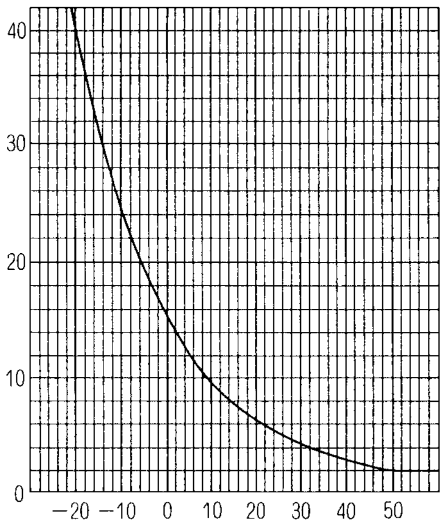
Resistance measurement point (connector)



Touch the probes of the tester or other instrument to the shaded areas to measure.

Temperature sensor resistance (graph)

Resistance value (kΩ)



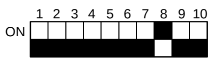
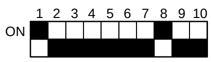
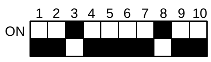
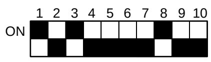
Temperature (°C)

Thermistor R_0=15 kΩ

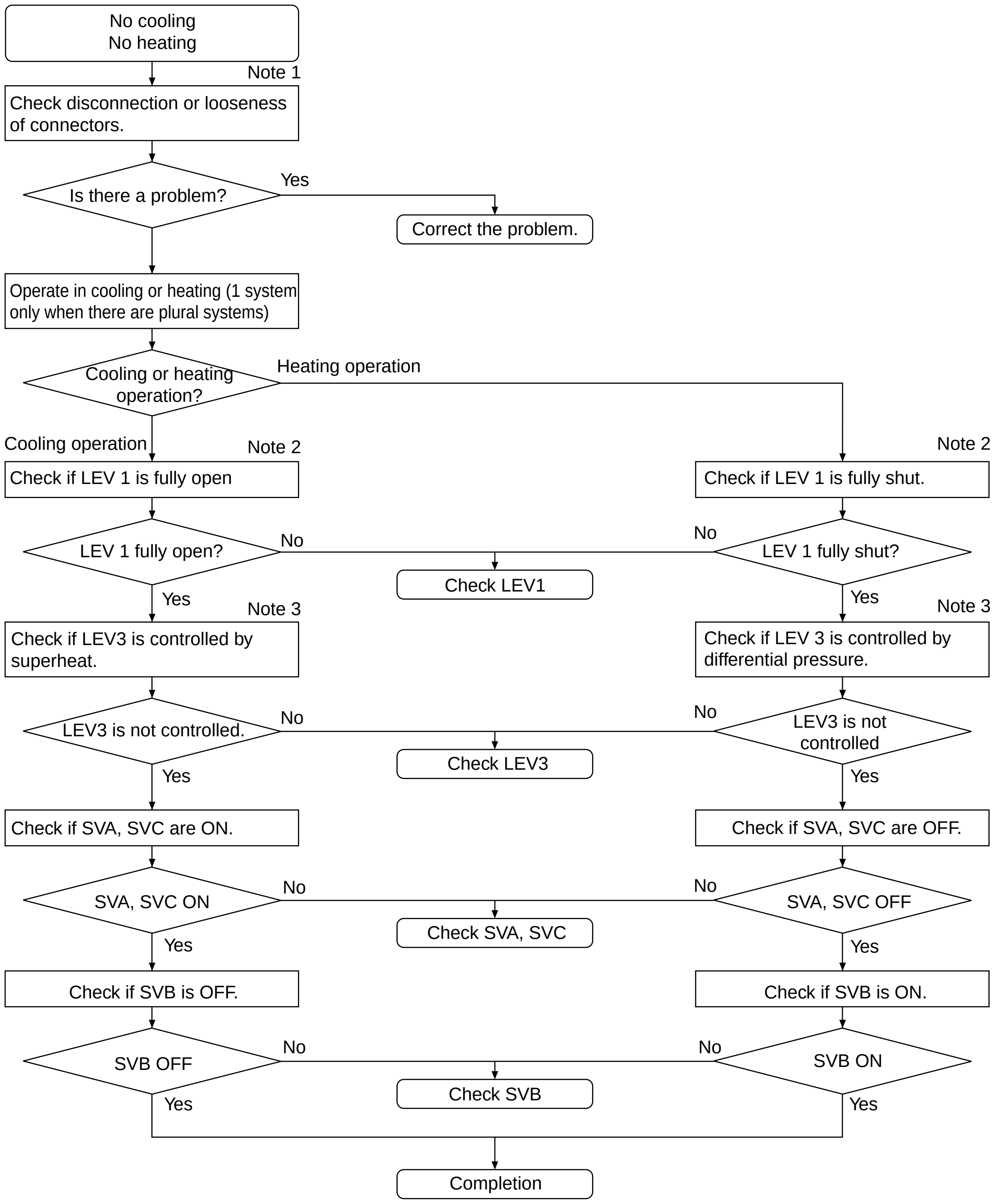
$$Rt=15\exp 3460\left\{\left(\frac{1}{273+t}-\frac{1}{273t}\right)\right\}$$

Note 4 :

- Check using LED monitor display switch (outdoor MAIN board SW1)

Measured Data	Signal	SW1 Setting
Liquid inlet temperature	TH11	
Bypass outlet temperature	TH12	
Bypass outlet temperature	TH15	
Bypass inlet temperature	TH16	

3) LEV, solenoid valve troubleshooting flow



① LEV

Note 1 :

- Symptoms of incorrect connection to BC controller LEV board

LEV No.	1	3	Cooling-only	Cooling-main	Heating-only	Heating-main
1)	1	3	Normal	←	←	←
2)	3	1	Insufficient cooling SH12 small, SC11 small SC16 small Branch piping SC small	Insufficient cooling, insufficient heating SH12 small, SC11 small SC16 large, Branch piping SC small △ PHM large	Heating indoor SC small △ PHM large	Insufficient cooling Heating indoor SC small △ PHM large

Improper installation is the same for ① and ②, so it is omitted here.

Note 2 : Method for checking LEV full open, full closed condition

① Check LEV full opening (pulse) using the LED monitor display (outdoor controller board SW1).
Full opened: 2000 pulses
Full closed: 60 pulses (LEV 1 may be greater than 60 during full heating operation.)

② With LEV full opened, check for pressure differential by measuring temperature of piping on both sides.

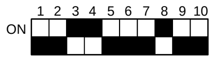
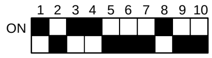
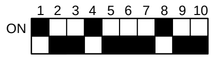
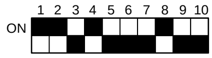
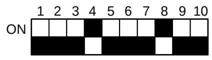
③ With LEV full closed, check for refrigerant noise.

Note 3 : Use the following table to determine opening due to LEV differential pressure control and superheat control.

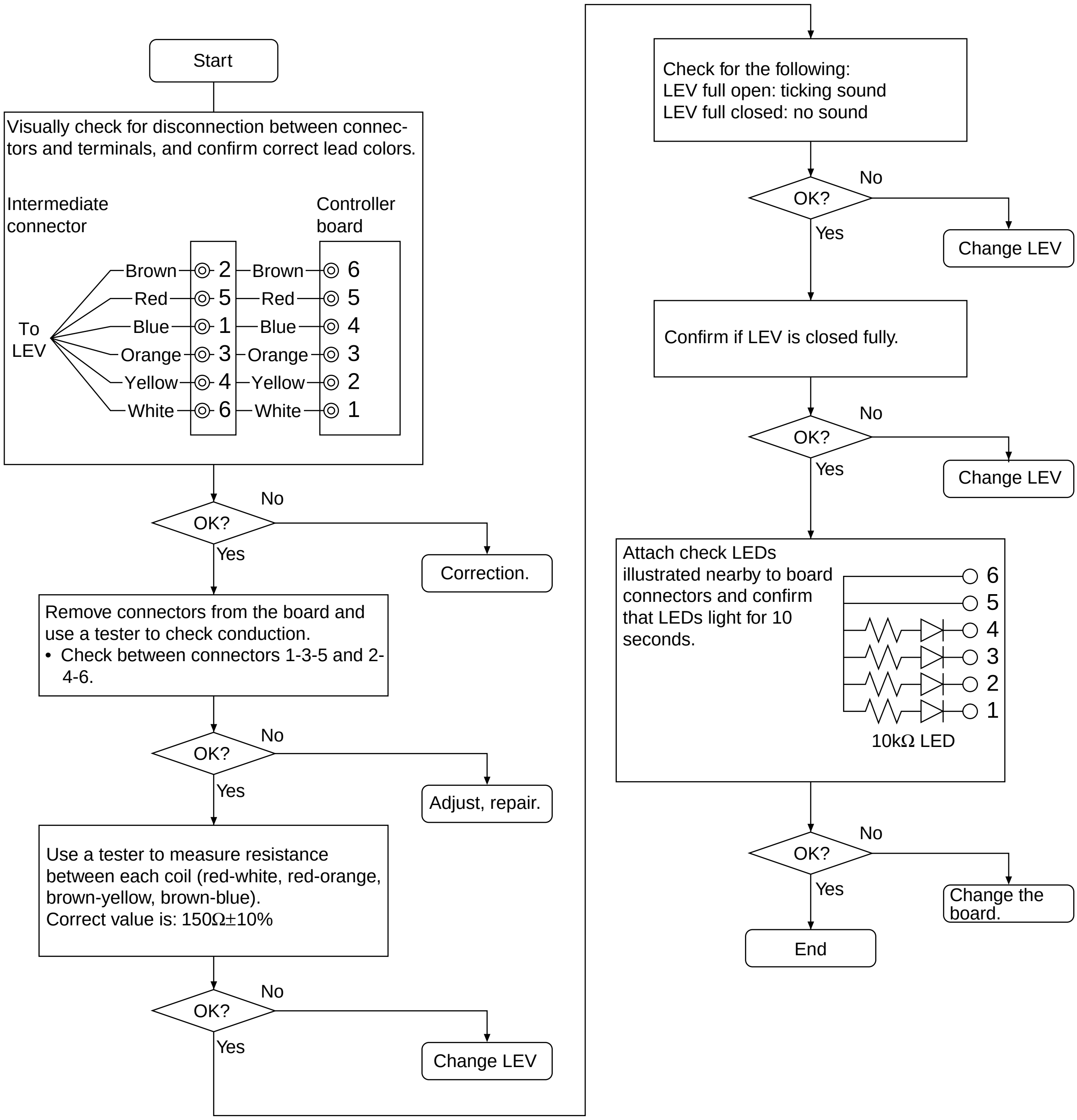
- BC controller LEV basic operation characteristics

Region	Failure mode	Operating mode	Description	Normal range
LEV1 pulse	Small	Heating-only Heating-main Cooling-main	High pressure (PS1) - medium pressure (PS3) is large.	0.20~0.34MPa
	Large		High pressure (PS1) - medium pressure (PS3) is small.	
LEV3 pulse	Small	Cooling-only Cooling-main	SH12 is large.	SH12<25
		Heating-only Heating-main	High pressure (PS1) - mid pressure (PS3) is small.	0.20~0.34MPa
	Large	Cooling-only Cooling-main	SC16 and SH12 are small.	SC16>6 SH12>5
		Heating-only Heating-main	High pressure (PS1) - mid pressure (PS3) is large.	0.20~0.34MPa

(Self-diagnostic monitor)

Measured data	Signal	Heat source unit MAIN board SW1 setting
LEV 1 pulse	–	
LEV 3 pulse	–	
BC controller bypass output superheat	SH12	
BC controller intermediate subcool	SC16	
BC controller liquid subcool	SC11	

(Solenoid Valve Troubleshooting Flow)



② Solenoid Valve